Sheikh Mohsin

+91 6005277810 • sheikh.mohsin@flame.edu.in
LinkedIn • GitHub • Portfolio

Profile Summary

Third-year B.Sc. Computer Science and Business Analytics student at FLAME University, Pune. Summer research on SSD cache eviction and spatial prefetching at IIT Ropar; a year of field research on Taraksi silver filigree; co-author of a Deccan Herald piece on education policy. Comfortable moving between systems code and qualitative fieldwork, and writing clearly about both.

Education

FLAME University, Pune Aug 2024 – Present
B.Sc. Computer Science & Business Analytics (Full Scholarship) Cumulative GPA: 7.65/10

- Modeled and verified concurrent and distributed systems with SAT solvers, temporal-logic model checking, and Alloy/TLA+ specifications
- Implemented and compared classification and clustering methods, using PCA to test how feature selection affected model behavior
- Term paper on globalization and India's LPG-era economic reforms (Public Policy coursework)
- Joined Dot Slash Club and completed interdisciplinary projects connecting computer science with Philosophy and Public Policy

Jawahar Navodaya Vidyalaya, Ganderbal 2017 – 2024
Senior Secondary Education (Merit Scholarship, Residential) Grades 6–12

- Led extracurricular activities and coordinated Atal Tinkering Labs; taught junior students to turn ideas into 3D-printed prototypes

Experience

IIT Ropar May 2026 – Present
Research Intern (Summer Internship Program) – SSD Systems

- Learned SSD storage systems from first principles under Prof. Venkata Kalyan Tavva and compared recency- and frequency-based cache eviction strategies in the SimpleSSD-Standalone 2.0 simulator as a foundation for an original performance study
- Designed and implemented a spatial prefetching approach that predicts nearby address-translation entries from how data is grouped in flash; found and fixed a cache-order bug that caused newly loaded entries to evict one another before use
- Investigated how ransomware write bursts interact with an SSD's internal cleanup process, identified a race that may destroy recoverable data before detection responds, and assessed whether a targeted defense could form a new research contribution

Discover India Program (DIP) Sept 2025 – April 2026
Research – Taraksi (Silver Filigree)

- Studied, within a 13-member research team, how Taraksi silver filigree moved from a household craft to a commercial practice
- Traced the craft's history and technique through literature, then designed field surveys and semi-structured interviews to capture how artisans describe the shift from craft to commerce
- Synthesized qualitative findings into academic documentation and cultural analysis

PartyHub (Remote) Jul 2025 – Aug 2025
QA Testing Intern

- Tested frontend and backend features before release, focusing on edge cases that could break real user flows rather than checking only the happy path
- Wrote reproducible bug reports with exact steps and expected-versus-actual behavior, giving developers a clear path from discovery to triage
- Checked API responses and cross-device behavior to expose integration issues before release

Goonj, Delhi May 2025 – Jul 2025
Research and Operations Intern

- Traced donation and redistribution data through the sorting pipeline, isolated stages that created backlog, and built structured datasets and visual summaries to support allocation decisions
- Tested data patterns against field-team observations before proposing process changes, separating patterns that reflected real field conditions from patterns caused by incomplete records

Projects

Cache Performance Benchmark Suite May 2026
C++, `perf_event_open`, RDTSCP Repository
Built a benchmarking harness that measured hardware clock-cycle costs across 16 data structures, using performance counters to isolate machine-level behavior from wall-clock noise. Compared contiguous (array-of-structs / struct-of-arrays) and pointer-based layouts across access patterns and data sizes, then automated collection, CSV export, and Chart.js visualization to make the results reproducible.

IPL Data Mining April 2026
Python, Data Analysis, Data Visualization Repository
Cleaned and explored IPL data to compare team and player patterns, turning match records into findings that could be inspected through visual summaries. Built a repeatable analysis workflow in Python, separating data preparation from interpretation so conclusions could be checked against the underlying records.

Personal Website & Portfolio 2025 – Present
HTML, CSS, JavaScript, Vercel Website
Designed and built a responsive personal website from scratch to present projects, research, and writing in one place. Deployed the site through GitHub and Vercel and kept the structure focused on making technical work easy to find and inspect.

Technical Skills

Programming & Tools: Python, C++, JavaScript, SQL, HTML/CSS, Git/GitHub, Linux, LaTeX, Markdown
Data & Systems: Data Structures & Algorithms, Data Analysis, Data Mining, Data Visualization, Statistical Reasoning, FTL/NAND flash internals, hardware performance profiling (`perf_event_open`, RDTSCP), basic AWS
Soft Skills: Technical writing, research communication, independent learning, analytical thinking, collaboration
Languages: English, Hindi, Urdu, Kashmiri

Leadership & Service

Karta Inter-University Workshops Dec 2025; Jul 2026
Technical Lead (Winter) & Operations Lead (Summer) Karta Initiative India Foundation

- Led technical operations during the five-day winter workshop, setting up and troubleshooting projectors, speakers, mixer amplifiers, microphones, and presentation displays so sessions ran without disruption
- Supported speakers and participants in real time by managing presentation changes, resolving equipment issues, and coordinating with the organizing team when problems arose
- Took broader responsibility for the summer workshop, coordinating housing, food, events, schedules, and related arrangements for students from Karta's university network and following each detail through to completion

FLAME University – Karta Initiative Aug 2024 – Present
University Leader

- Served as a peer representative for a cohort of more than 40 Karta Scholars, running weekly Sabhas that gave students a regular space to share updates, raise concerns, and agree on next steps
- Onboarded new scholars and carried information between the cohort and Karta Initiative India Foundation, keeping students informed about requirements, opportunities, and support
- Collected grades, attendance, banking details, and other records, while checking in with students who became disengaged and escalating larger concerns to the assigned mentor

Writing & Publications

Deccan Herald: Co-authored article on education policy and technology – "Lost in Instruction"
Kashmir Poetics: Contributed reflective and literary writing, including a published poem – kashmirpoetics.com
Comfortable synthesizing research material and communicating technical ideas to diverse audiences.